

SANMATI RAJIV SAWALWADE

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EDUCATION

Northeastern University, Boston, MA | M.S. Information Systems | GPA 4.0/4.0 | Expected May 2027

Coursework: Machine Learning, Data Science Engineering, Algorithms & Data Structures, Application Engineering, Business Analysis

Dr. D. Y. Patil College of Engineering | Bachelor of Engineering | Mar 2022

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, Java, JavaScript/TypeScript, C, C++, R

ML / Data Science: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face, Transformers, LLMs (GPT, Claude API), CNNs, SHAP

Data Engineering: Apache Spark, PySpark, Airflow, ETL Pipelines, Batch Processing, Data Warehousing, Data Wrangling

Databases: PostgreSQL, BigQuery, MongoDB, Relational Databases, Vector DBs (Pinecone, Weaviate)

Cloud & DevOps: AWS, Docker, REST APIs, FastAPI, CI/CD, Git, GitHub Actions, Production ML Pipelines, Model Monitoring

Analytics & BI: Tableau, Power BI, Mixpanel, GA4, A/B Testing, Bayesian Methods, Cohort & Funnel Analysis, Hypothesis Testing

PROFESSIONAL EXPERIENCE

Audifytech

Software Engineer | Sep 2022 to Dec 2023

- Built and deployed production ML classification model (**0.82 AUC-ROC**) in Python/Scikit-learn serving **1M+ users**; designed RESTful API endpoints for model inference and ran a Bayesian A/B testing framework across 15+ experiments, lifting **D7 retention 7%** and protecting **\$140K+ ARR**
- Engineered batch data pipeline with Airflow workflow orchestration, clustering 100K+ users (K-Means + DBSCAN) into behavioral segments; built feature embeddings and a retrieval layer powering a recommendation engine with **23% precision lift**
- Designed production model monitoring with structured logging, drift tracking, latency profiling, and SHAP feature importance across inference endpoints at **sub-100ms p95**

Gameskraft (\$2.3B Unicorn)

Software Engineer Intern | Sep 2021 to Dec 2021

- Built ensemble ML models (logistic regression + random forests) on a **10M+ event dataset** using Python and SQL; surfaced key LTV drivers, reducing **first-week churn 12%** across 3 products serving **10M+ users**
- Executed hypothesis testing and survival analysis to validate ML logic across 15+ production releases; collaborated with cross-functional product, data, and engineering teams, achieving **zero post-release regressions**

Pocket52

Software Engineer Intern | Nov 2020 to Aug 2021

- Mined 100K+ onboarding journeys via SQL funnel analysis; identified 3 statistically significant drop-offs and partnered on a data-driven UX redesign that lifted **conversion 4pp**, unlocking **\$120K+ incremental MRR** in 60 days; shipped automated BI dashboards (Python, SQL, Tableau) that eliminated **80% of manual reporting**

Dota2booststriker ([Link](#))

Founder & Full-Stack Engineer | Jul 2019 to Oct 2020

- Founded full-stack web platform; architected backend with Docker, REST APIs, and payment integration, scaling to **\$16K+ ARR** by Month 3 (4.8/5.0 rating, 95%+ retention, 200+ transactions)

PROJECTS

COVID-19 CT Scan Classification | Medical Imaging Deep Learning ([Link](#))

Python, TensorFlow/Keras, EfficientNet, Transfer Learning, Scikit-learn

- Built deep learning classifier on the **COVIDx CT** dataset triaging chest CT scans into COVID-19 / Pneumonia / Normal; fine-tuned EfficientNetB0 via two-stage transfer learning with class-weighted loss, achieving **93.4% test accuracy** (0.91 to 0.97 per-class F1) and validated robustness with 5-fold stratified CV (86.7% $\pm$ 1.2%)

NLP Document Retrieval & LLM Inference System

Python, LangChain, Pinecone, Transformers, FastAPI, Docker

- Designed multi-stage RAG pipeline with Transformer embeddings, semantic chunking, and Pinecone vector search for **sub-200ms retrieval** over **50K+ documents**, achieving **89% answer relevance** on NLP eval benchmarks; deployed via Docker + FastAPI with LangChain multi-agent orchestration and observability logging

FirstApply | Real-Time Internship Intelligence Platform ([Link](#))

Python, Node.js, PostgreSQL, Claude API, Playwright, BeautifulSoup, FastAPI, Docker

- Architected full-stack job intelligence platform scraping **6+ live job boards** (LinkedIn, Indeed, Greenhouse, YC, HackerNews) in real-time using Playwright and BeautifulSoup, aggregating **2,600+ roles** with deduplication and structured PostgreSQL storage
- Built NLP-powered resume scoring engine using the **Claude API** to parse resumes, extract skill/experience vectors, and score each job 0 to 100 against up to 5 candidate profiles for personalized job-fit ranking
- Engineered real-time alerting pipeline with near-zero notification latency; early applicants are **3x more likely** to receive callbacks

ACHIEVEMENTS & PUBLICATION

- 3 production ML models at scale (1M+ users):** 7% retention lift, 12% churn reduction, 4pp conversion lift, \$200K+ cumulative impact | Google Play Users' Choice Nominee 2023 (*Football World*) ([Link](#)) | Published in IJET Vol. 8 (2022)
- Published Researcher** | "Waste Reduction Using Value Stream Mapping" | IJET, Vol. 8, 2022 | Applied VSM process optimization to reduce **lead time 27%**, eliminating bottlenecks through cross-functional coordination